

Cryptography Basics

Cryptography is the **science of securing information** by converting it into an unreadable format to protect it from unauthorized access. It ensures **confidentiality, integrity, authenticity, and non-repudiation** of data.

Encryption : used to protect data from unauthorized access by converting it into an unreadable format.

Types of Encryption: Symmetric (AES) vs. Asymmetric (RSA).

 Key Differences: Symmetric (AES) vs. Asymmetric (RSA)		
Feature	Symmetric (AES)	Asymmetric (RSA)
Keys Used	1 (Secret Key)	2 (Public & Private Keys)
Speed	Fast	Slow
Security	Less secure if the key is exposed	More secure (private key is never shared)
Best Used For	Large data encryption (files, databases)	Secure key exchange, authentication, digital signatures
Example Algorithms	AES, DES, 3DES	RSA, ECC, Diffie-Hellman

Hashing algorithms (MD5, SHA-256) and digital signatures.

Hashing

- a one way process that converts data into a fixed-length string (hash)
- it ensures integrity (cannot get the original data from the hash)

1. MD5 (Message Digest Algorithm 5)

- **Output size:** 128-bit (32-character hexadecimal)
- **Fast but weak** – Collisions can be found (not secure for cryptography)
- **Used for:** Non-cryptographic checksums (file integrity, non-sensitive data verification)

MD5 is NOT recommended for security purposes due to known vulnerabilities.

2. SHA-256 (Secure Hash Algorithm-256)

- **Output size:** 256-bit (64-character hexadecimal)
- **More secure than MD5** – Stronger against collisions
- **Used for:** Cryptographic security, password hashing, digital signatures, blockchain

3. Digital Signatures

A digital signature is a cryptographic technique used to **verify authenticity and integrity** of a message or file.

♦ How Digital Signatures Work

- 1 Sender creates a hash of the message.
- 2 Sender encrypts the hash with their **private key** (this is the digital signature).
- 3 Receiver decrypts the signature using the sender's **public key**.
- 4 Receiver hashes the received message and compares it with the decrypted hash.
 - If **both hashes match**, the message is authentic.
 - If **hashes don't match**, the message was **tampered with**.

Encrypt & Decrypt a file using OpenSSL:

1. AES-256 File Encryption & Decryption

```
aditi@aditi-IdeaPad-3-15IAU7:~$ echo "This is a test file for encryption" > plaintext.txt
aditi@aditi-IdeaPad-3-15IAU7:~$ openssl enc -aes-256-cbc -salt -in plaintext.txt -out encrypted.bin -pbkdf2
enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:
aditi@aditi-IdeaPad-3-15IAU7:~$ openssl enc -d -aes-256-cbc -in encrypted.bin -out decrypted.txt -pbkdf2
enter AES-256-CBC decryption password:
aditi@aditi-IdeaPad-3-15IAU7:~$
```

2. RSA Key Pair Generation (Asymmetric Encryption)

[illegible]

3. Verify hash integrity of a file using SHA-256.

SHA-256 hashing ensures that a file has **not been modified** by comparing its hash before and after transfer.

```
aditi@aditi-IdeaPad-3-15IAU7:~$ openssl dgst -sha256 plaintext.txt
SHA2-256(plaintext.txt)= dcdab874cad8a939f6a91c5d022c6be08f4c8d307024535352b22e3
e98a1c74e
aditi@aditi-IdeaPad-3-15IAU7:~$ openssl dgst -sha256 plaintext.txt > hash.txt
aditi@aditi-IdeaPad-3-15IAU7:~$ openssl dgst -sha256 plaintext.txt
SHA2-256(plaintext.txt)= dcdab874cad8a939f6a91c5d022c6be08f4c8d307024535352b22e3
e98a1c74e
aditi@aditi-IdeaPad-3-15IAU7:~$ cat hash.txt
SHA2-256(plaintext.txt)= dcdab874cad8a939f6a91c5d022c6be08f4c8d307024535352b22e3
e98a1c74e
aditi@aditi-IdeaPad-3-15IAU7:~$
```

If the hashes match → The file has not changed.

```
aditi@aditi-IdeaPad-3-15IAU7:~$ nano verify_hash.sh
aditi@aditi-IdeaPad-3-15IAU7:~$ chmod +x verify_hash.sh
aditi@aditi-IdeaPad-3-15IAU7:~$ ./verify_hash.sh
❌ WARNING! File was altered or corrupted.
aditi@aditi-IdeaPad-3-15IAU7:~$
```

If the hashes are different → The file was altered or corrupted.

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~Task 2 Week 5